Data Mining Techniques for the Analysis and Prediction in Bioenergy Yield

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***Abstract*—Unconstructed data mining is challenging task in current data mining research domain. Important information than text since it widely used and content very rich information, such as experiment results and parameters setting. This paper examines the large number of tables that existed in the bioenergy field and propose a framework for automatically acquiring data from papers in PDF format and outputs the table information. To evaluate the performance of our automatic data acquiring method, we compare four existing software. 110 tables have been used for evaluation and the final overall score of our method is 90.54 which shows our method has the best performance in terms of position accuracy. This result demonstrates that our approach is promising and scalable. This work also provides a foundation of data processing for bioenergy.**

Keywords—Table mining; Information extraction; Biochar; Automatic system

I. Introduction

In the field of bioenergy, gasification has received increasing attention because it has a great potential to tackle a triple crisis of increasing energy demand, climate change and waste pileup [1]. With the increasing experimental data, it is valuable to analysis those experiment data to avoid repeating experiments and provide prediction for the bioenergy yield through data mining techniques. Since each experiment in bioenergy will last more than 1 month. However, it is a daunting task to retrieve these considerable data manually. Although many existing text mining tools can operate on papers, most of them only deal with structured text which means they usually do not process tabular data, even if it does, it does not work very well.

Tabular data is easy to be understood by humans, but it can be a complex task for computer [2]. Particularly, for tabular data that exists in PDF, this extraction and reconstruction becomes much more challenging task due to PDF documents are difficult to convert inerrably.

Presenting results in tables is very popular and convenient way used in bioenergy research domain. Extracting the tables automatically and correctly will bring the possibility of using data mining processes to help researchers to design bioenergy experiments. Since the literature will provide a good starting point for researcher to bioenergy yield predication, furthermore, to help them designing new experiments. Therefore, a process-based and automated system needs to be designed to conducting data acquiring from tables.

In this paper, we demonstrate a framework (as shown in Figure 1) that can automatically extracts information from tables and stores the data as CSV format. Paper fetching is based on web crawler and DOI of a given paper. After the relative papers downloaded, they will be converted into HTML format using open resource software pdfminer [3]. The auto-detection tool will locate the paper position and extract the information. The final information can be saved as excel format and csv format with user's choice. Although focusing on the papers related to bioenergy yield, we believe our framework can be used in other field, such as healthcare and social science.

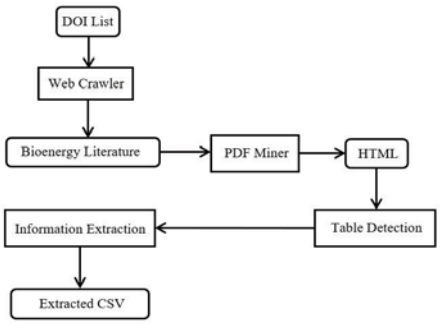

Fig. 1. Framework

II. Background

Hurst [4] started the pioneer work using text mining method. Tables have been modeled and the whole task was briefly divided table mining into table recognition, functional analysis and information extraction respectively. Moreover, he developed his table mining engine and describe it as TabPro.

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For table detection, many researchers have implemented it in different ways. Kieninger and Strieder (1999) worked on table recognition from PDF, HTML and ASCII documents by OCR (Optical Character Recognition) [5]. They developed "T-Recs" system which could read paper scanned documents.

Yildiz et al. (2005) built a tool named pdf2table [6]. This tool is based on pdf2html tool. It can convert PDF to XML, so the problem is to extract tables from XML. This method also has some limitations, that is, it may not be able to extract all the tables correctly because PDF files are difficult to convert without error. To deal with this limitation, a graphical interface that allows users to determine the accuracy of information acquired is designed.

Ng et al. (1999) deal with table detection by two machine learning algorithms C4, 5 decision trees and ANN (Artificial Neural Networks) [7]. Similar machine learning algorithms are used by JW et al (2008) [8] and Silva who applied several algorithms on table mining tasks like SVM and heuristics [10].

Constantin et al. (2013) construct a rule-based system [9]. The input of the system is a PDF document and the output is an XML document. Although their system manifests a successful performance in recognizing tables, converting tables in PDF document into XML file that can be parsed by computer properly does not perform well.

Hearst et al. (2007) and Divoli et al. (2010) built a system called "BioText Search" which could extract tables from PubMed Central database of biomedical literature and based on same data sets, Xu et al. (2008) developed "Yale image finder" which applied OCR to read text from images in PubMed Central articles [11]. So far, most of the research have been in the biomedical field and any work implemented on extracting tabular data in bioenergy literature are ignored.

III. METHODOLOGY

Our goal is to extract tabular data from a pool of papers in the field of bioenergy. For simplicity, we focus on the three-line table (like Figure 2) which is most papers in this field. There are few or no ordinary tables (like Figure 3). This means that each piece of data cannot be stored in a data cell and compared with the ordinary table, three-line table has more complex characteristics for computer recognition.

(a) Wood biochar

	Highest treatment temperature (°C)			
Residence time (min)	300	450	600	750
10	–	4	196	128
60	6	23	127	–

Fig. 2. Example of Three-line table

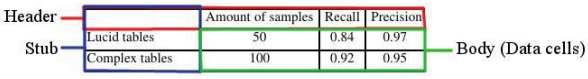

Fig. 3. Example of ordinary table

The method has been proposed could be divided into three parts: firstly, convert PDF papers to HTML files by PDFMiner [12] based on python in batch. Secondly, find all tables sections in the HTML according to rules that will be demonstrated in subsection A (Table content extraction). Finally, the corresponding information is extracted through our method, and the result is presented as the format of CSV.

A. Table content extraction

Because PDF cannot be converted to HTML without lossless, the converted HTML file is not in standard HTML format. Sometimes there is a leading tag but no trailing tag. In other words, if the HTML document is forced to be converted to PDF, the original layout and structure will not be restored correctly, and this scenario is also true for tables in paper. Figure 4 shows a table representation in a standard HTML document. It implies that if we want to locate the table of an HTML document, it only need to search the tag "table"(<table ...>...</table>) and the information in header is stored in tag "thead"(<thead>...</thead>) while table body is in tag "tbody"(<tbody>...</tbody>).

```
▼<table class="wikitable sortable jquery-tablesorter">
  ▼<thead>
    ▶<tr>…</tr>
    </thead>
  ▼<tbody>
    ▶<tr>…</tr>
    ▶<tr>…</tr>
    ▶<tr>…</tr>
    ▶<tr>…</tr>
    ▶<tr>…</tr>
    ▶<tr>…</tr>
    ▶<tr>…</tr>
    ▶<tr>…</tr>
    ▶<tr>…</tr>
    ▶<tr>…</tr>
    </tbody>
    <tfoot></tfoot>
  </table>
```

Fig. 4. Example of table in a standard HTML

Therefore, if the table in an HTML document is in standard form, it will be easy to locate the tabular data. However, the HTML documents converted from Pdfminer were less friendly. Figure 5 shows that the HTML that is converted by Pdfminer. We can see that the "table" field no longer appears in the tag but sneaked into as a piece of text. Not only that, the header and body data in the table all exist as a single section in . So, we propose a way of finding tables.

Fig. 5. Example of table in output of Pdfminer

Through the observation of data of table retrieved from HTML files (see Figure 6), we can see each table begins with a "Table" string, so we can use it as the starting point of extracting table content. However, usually the table will also be referenced in the body of the article, so this string cannot be used as the only indicator. Through the observation and comparison of a large number of HTML files generated from articles, we found we could use the specification description ("font-size: 7px", etc.) to tell their difference as there is no

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specification description in the main body. Therefore, we can write a regular expression to uniquely match the field including "Table" and the specification description as the beginning of the table. As for the end of table content, we could use coordinate to locate. We can see each entry in the table has four parameters to describe position: "left", "top", "width", "height". Through the consulting literature [13], we know that "left", and "top", represent the horizontal and vertical coordinates of the upper left corner of the entry in the table, and "width", and "height", are exactly the same as they are literally, representing the width and height of the entry so the horizontal and vertical coordinates of the last element must be the largest. Therefore, we could use text search to find all the coordinates until we find the biggest coordinate as the end. The algorithm flow chart is shown in Figure 6.

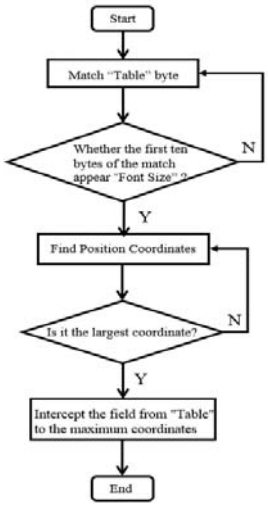

Fig. 6. Example of table in output of Pdfminer

B. Relative position determination

1) Extraction of coordinates:

The extraction of coordinates can be achieved by defining regular expressions. As mentioned earlier, there are four indicators are used: "left", "top", "width", "height", with "left" and "top" representing the horizontal and vertical coordinate of top left corner. Therefore, we can only use these two coordinates to determine their position in the table uniquely. Specifically, we can define the regular expression as following: "\bleft: (\d2, 4)px$","\btop:(\d2,4}px". Through experimental observation, regular expressions can accomplish coordinate extraction tasks very well.

2) Coordinate integration:

After extracting the coordinates, we can rearrange the extracted content in the Excel according to the coordinates. For any two elements in the same column, their "left" value should be the same, but due to some table layout problems, their left value will have a slight difference. In this case, we should treat them as the same column instead of different columns. The situation is the same for the rows. To solve this problem, we set a threshold to integrate different entries in different columns or rows with similar coordinates to a single column or row. Generally, papers are written by word or latex and then transcoded to pdf. For finding a proper threshold, we have studied the encoding of the latex and word to tables. We also observed and compared the arrangement of several academic papers, and we decide to choose the threshold of horizontal coordinate to be 8px, and the threshold of vertical coordinate to be 10px.

C. Mapping a single piece of information to a location

After the above steps, what we get is the absolute position of a section (represented in HTML by a $<div>$) and the relative position of each section. In fact, the number of information under each section is uncertain. There may be only one data (shown in Figure 7) or multiple data (shown in Figure 8). Therefore, the next thing we need to do is position the individual or multiple pieces of information under a section.

```
<div style="position:absolute; border: textbox 1px solid; writing-mode:lr-tb; left:41px; top:1826px; width:34px; height:7px;">
<span style="font-family: b'EAPPNF+AdvGulliv-R'; font-size:7px">Parameters<br></span>
</div>

<div style="position:absolute; border: textbox 1px solid; writing-mode:lr-tb; left:133px; top:1826px; width:29px; height:7px;">
<span style="font-family: b'EAPPNF+AdvGulliv-R'; font-size:7px">Feedstock<br></span>
</div>
```

Fig. 7. A section conclude only one data

```
<div style="position:absolute; border: textbox 1px solid; writing-mode:lr-tb; left:41px; top:1946px; width:11px; height:33px;">
<span style="font-family: b'EAPPNF+AdvGulliv-R'; font-size:7px">200
<br>400
<br>600
<br>800
<br></span></div>
```

Fig. 8. A section conclude multiple data

We analyzed many tables and found that the general rule is that HTML documents tend to reduce a series of columns to a single section with little or no reduction of multiple data in a row to a single section. The main reason is that under a particular section, an HTML document can use <br> to distinguish whether there is a newline between one piece of data and the next, rather than whether they belong to different columns and actually, is used to distinguish between columns most of the time. Then we can make the following rules. Suppose the relative position of section A is [x, y] (where x is the number of rows and y is the number of columns), and the relative position of the next section B is [x, y+1]. There are N <br> below section A, which means that there are N tabular data below the A section. Therefore, the first step is to extract the data according to the number and position of <br>. In the premise of tag, we still use this rule, that is, when <br> appears, we think that the tag is separate two data, otherwise no matter how many tags appear, we think it is only one data. For example, see Figure 9, even if there are 3 , but in fact it's just one data (i.e., Pyrolysis temperatures). Then it could generate new relative position coordinates between [x, y] and [x, y+1]. In the case of N $
$ in A section, the position that we generate is [x+1, y], [x+2, y], [x+3, y] … [x+N-1, y]. With this rule, we end up with coordinate information that is consistent with the amount of data we have, and then it can write to an Excel file.

```
<div style="position:absolute; border: textbox 1px solid; writing-mode:lr-tb; left:41px; top:1933px; width:83px; height:8px;">
<span style="font-family: b'EAPPNF+AdvGulliv-R'; font-size:7px">Pyrolysis temperatures (<</span>
<span style="font-family: b'EBAACI+AdvPSSym'; font-size:8px">°</span>
<span style="font-family: b'EAPPNF+AdvGulliv-R'; font-size:7px">C)
<br></span></div>
```

Fig. 9. One data with several tag "span

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IV. Evaluation Methodology

To measure the performance of our method, a few criterions are introduced for comparison. We chose these criteria because the set of criteria reflects whether a table is integrally and correctly extracted. Our method is compared with four existing methods (only for table extraction goodness) by these criterions.

a) Position accuracy: Using it to measure whether positions of data extracted from pdf files are correct. If extracted data is not in the correct row or column, it will be recorded as an error. But a row or column consisted of blank space will not be considered as an error. This is because a blank row or column is not likely to affect readability of table extracted. It is calculated using the equation (1),

$$PA = \frac{Correct\ positions}{The\ total\ amount\ of\ extracted\ data} \tag{1}$$

b) Integrity: This is created to measure whether all useful data are successfully extracted. A formula to calculate it is given as equation (2),

$$I = \frac{Data\ extracted}{The\ total\ amount\ of\ data} \tag{2}$$

c) Precision: This term is applied to estimate the accuracy of data inside extracted table. It is calculated by the equation (3),

$$P = \frac{Correct\ extracted\ data}{The\ total\ amount\ of\ extracted\ data} \tag{3}$$

There are 100 different tables from the paper collection, and the criterions of five of them are shown in the Table I. Three of the Numbers are PA, I, P from left to right.

TABLE I. Example of several criterions for 5 table

PDFTables	PDFtoXLS	SimplyPDF	Tabula	Our Method
0.92/1/1	0.857/1/1	1/1/1	078/1/1	1/1/1
0.65/1/1	0.26/0.74/1	1/1/1	Failed	0.37/0.41/0.74
…	…	…	…	…
0.82/1/1	1/1/1	0.53/1/1	0.74/1/1	0.95/1/1
1/1/1	Failed	1/1/1	0.706/1/1	0.941/1/1
1/1/1	1/1/1	1/0.898/1	Failed	1/1/1

Finally, combining these three indicators, we define a scoring rule, and the final scoring of these five methods is shown in Table II.

$$E = log_{10}(10 \times PA \times I \times P) \tag{4}$$

TABLE II. Example of several criterions for 5 table

	PDFTables	PDFtoXLS	SimplyPDF	Tabula	Our Method
E	0.928	0.7721	0.7569	0.7427	0.9054

From the data, each method performs better in specific situations and can handle problems which other methods cannot. In terms of position accuracy, our method has the best performance. Tabula does the best job in collecting complete data and keeping precision but performs not so well in position accuracy. This can cause serious problem in table reconstruction. Both PDFTable and our method can locate data more correctly in the table extracted than other methods. By calculating the value of E, PDFTable possesses the best performance taking all the three terms into consideration. But in general, our method has the best availability, which means it can perform well under most situations without failure.

V. Conclusion

Data acquisition using table is not a widely studied topic, but it is of great research value in bioenergy. In this paper, we present a method for extracting useful information from tables and we design a system that can automatically obtain papers, convert PDF documents and extract tables. Among them, we pay more attention to the part of table recognition and extraction. Finally, we design a set of evaluation system (including position accuracy, integrity and precision). We then compare our method with some existing table extraction software, and it is shown an excellent performance.

From the performance of our result, we are convinced that our current work can be promising. Its main purpose is to improve the research efficiency of researchers working in the field of bioenergy. However, it is also scalable, which means that the same approach can be applied to other fields. Although our method is not perfect and there is room for improvement, we believe that it is of practical value to researchers.

Avalability

Our code and example are available on GitHub (https://github.com/text-mining-project/Table-information-extraction). Further work applied to others field will be developed.

References

[1] C. Higman, "Gasification", *Combustion Engineering Issues for Solid Fuel Systems, Academic Press*, 423-468, 2008. J. Clerk Maxwell, "A Treatise on Electricity and Magnetism", *3rd ed., Oxford: Clarendon*, vol. 2, pp.68-73, 1892.

[2] A. Tengli, Y. Yang, N. L.Ma, "Learning table extraction from examples", *Proceedings of the 20th international conference on Computational Linguistics. Association for Computational Linguistics*, pp. 987, 2004.

[3] Y. Shinyama, "PDFMiner: Python PDF parser and analyzer (2010)", *Cited on,* vol. 13, 2015.

[4] M F. Hurst, "The interpretation of tables in texts", 2000.

[5] T. G. Kieninger, B. Strieder, "T-recs table recognition and validation approach", *//AAAI Fall Symposium on Using Layout for the Generation, Understanding and Retrieval of Documents*, 1999.

[6] B. Yildiz, K. Kaiser, S. Miksch, "pdf2table: A method to extract table information from pdf files", *//IICAI,* 1773-1785, 2005.

[7] H. T. Ng, C. Y. Lim, J. L. T.Koo, "Learning to recognize tables in free text", *Proceedings of the 37th annual meeting of the Association for Computational Linguistics on Computational Linguistics, Association for Computational Linguistics*, pp. 443-450, 1999.

[8] J. W. Son, J. A. Lee, S. B. Park, et al., "Discriminating meaningful web tables from decorative tables using a composite kernel",*//2008 IEEE/WIC/ACM International Conference on Web Intelligence and Intelligent Agent Technology*, *IEEE*, vol. 1, pp. 368-371, 2008.

[9] A. Constantin, "Automatic structure and keyphrase analysis of scientific publications", *The University of Manchester (United Kingdom)*, 2014.

[10] N. Milosevic, G. Nenadic, "Table mining and data curation from biomedical literature", *End of first year PhD report,* 2014.

[11] S. Xu, J. McCusker, M. Krauthammer, Yale "Image Finder (YIF): a new search engine for retrieving biomedical images", *Bioinformatics,* vol. 24, no. 17, pp. 1968-1970, 2008.

[12] Y. Shinyama, "PDFMiner: Python PDF parser and analyzer (2010)", *Cited on*, vol. 13, 2015

[13] B. Tim, et al., "Extensible markup language (XML) 1.0", pp. 2000-10, 2000.

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